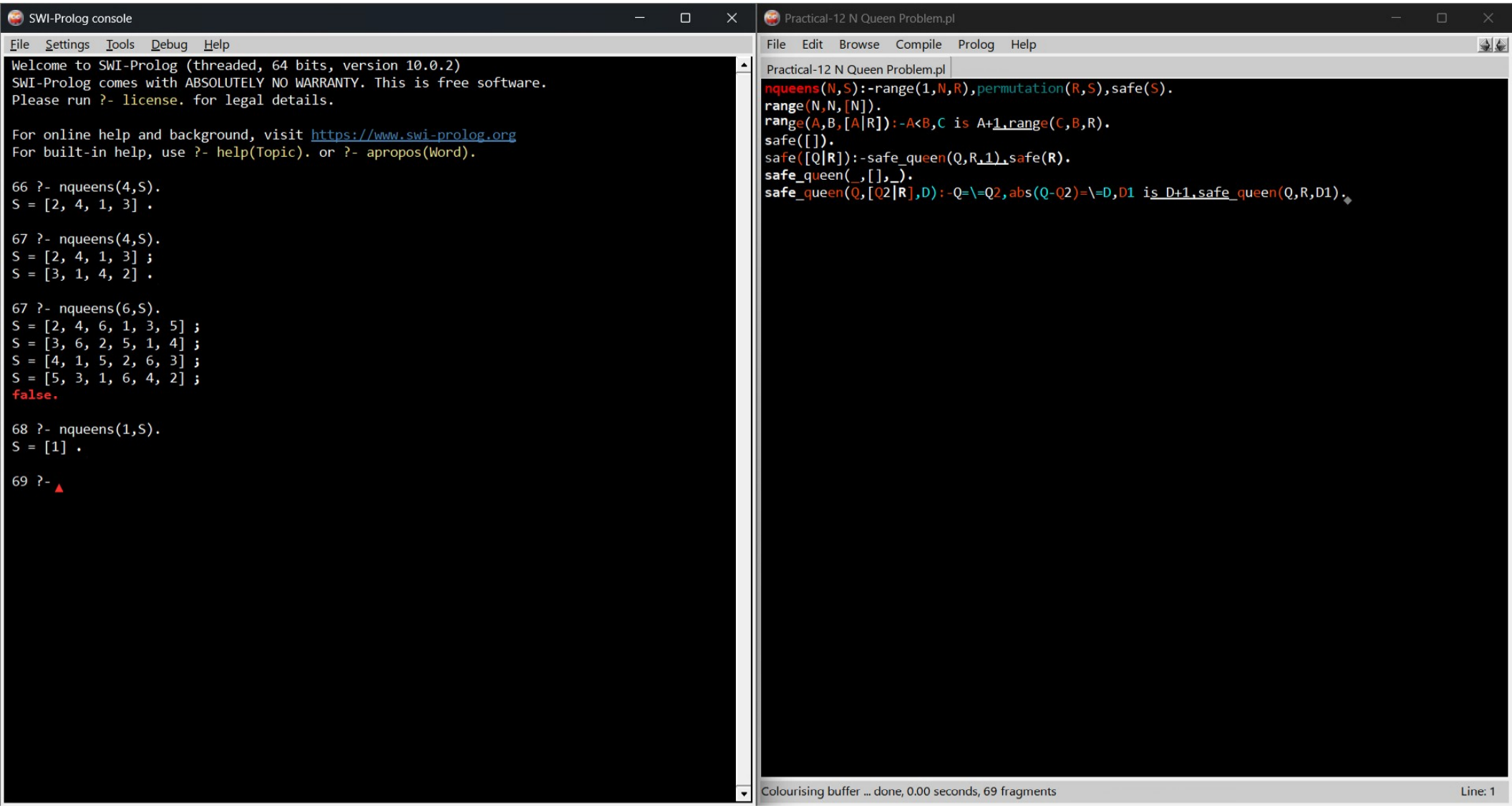
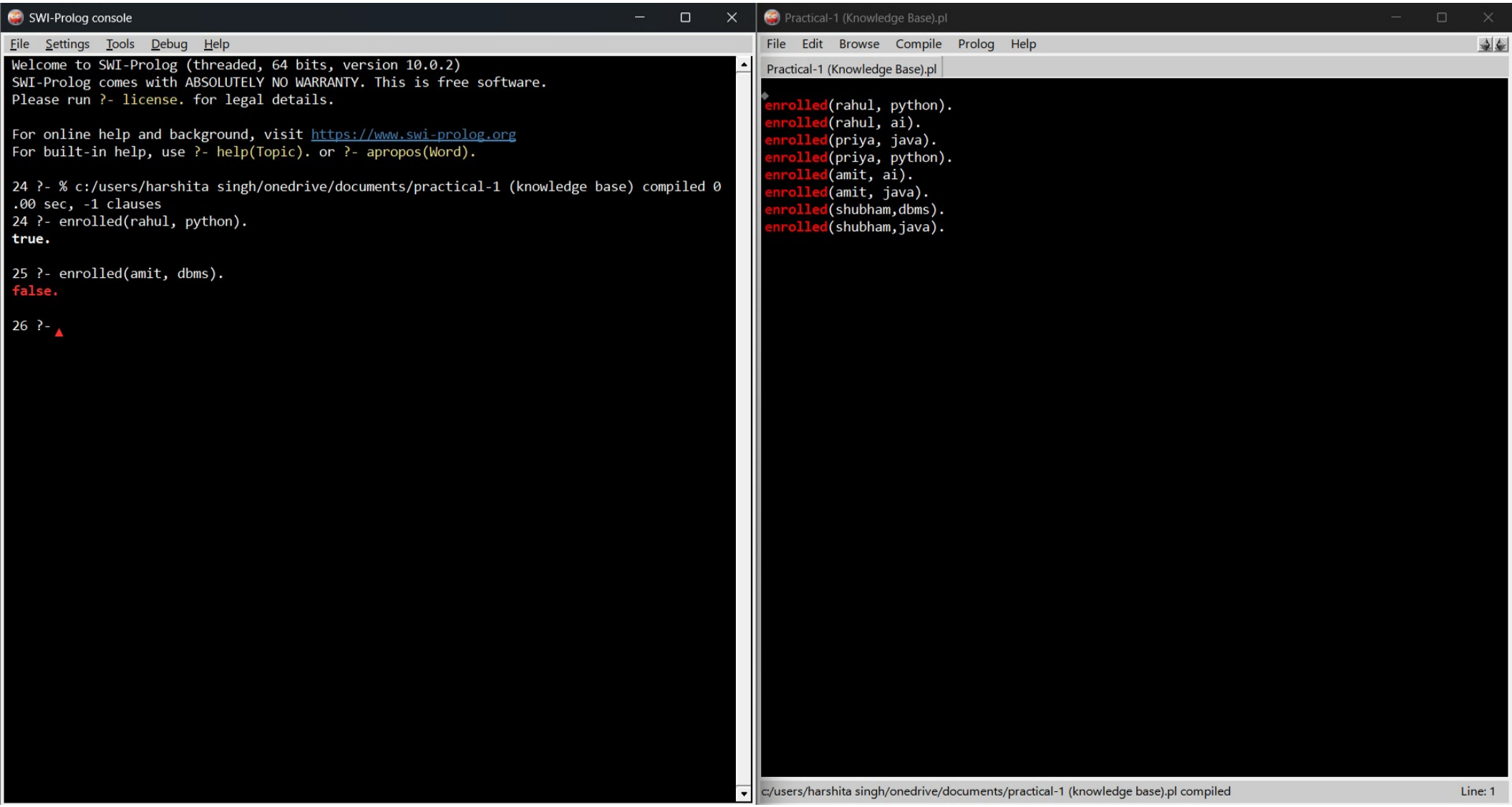
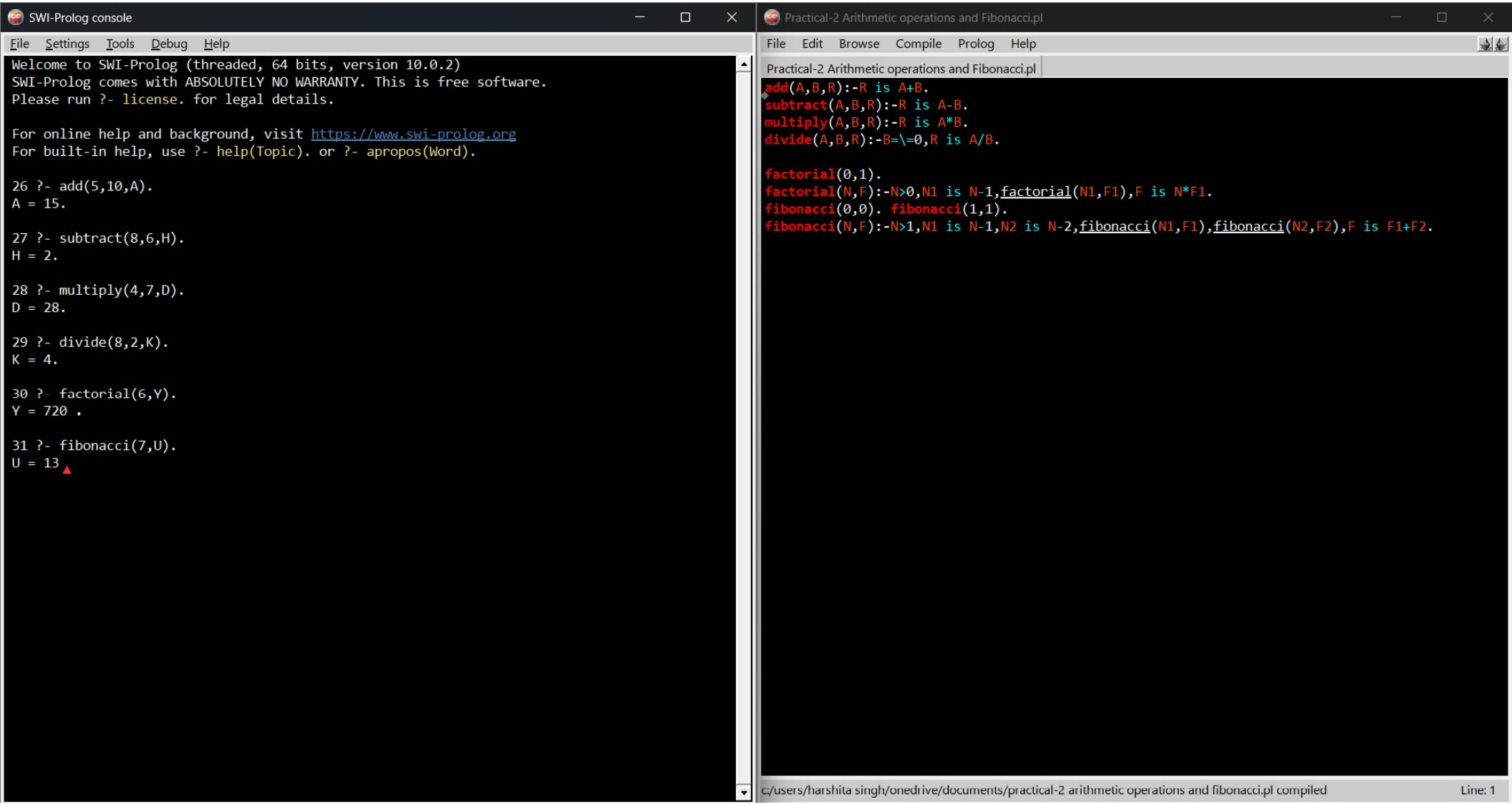

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For built-in help, use ?- help(Topic). or ?- apropos(Word).
66 ?- nqueens(4,S).
S = [2, 4, 1, 3] .
67 ?- nqueens(4,S).
S = [2, 4, 1, 3] ;
S = [3, 1, 4, 2] .
67 ?- nqueens(6,S).
S = [2, 4, 6, 1, 3, 5] ;
S = [3, 6, 2, 5, 1, 4] ;
S = [4, 1, 5, 2, 6, 3] ;
S = [5, 3, 1, 6, 4, 2] ;
false.
68 ?- nqueens(1,S).
S = [1] .
69 ?-
Practical-12 N Queen Problem.pl
File Edit Browse Compile Prolog Help
Practical-12 N Queen Problem.pl
nqueens(N,S):-range(1,N,R),permutation(R,S),safe(S).
range(N,N,[N]).
range(A,B,[A|R]):-A<B,C is A+1,range(C,B,R).
safe([]).
safe([Q|R]):-safe_queen(Q,R,1),safe(R).
safe_queen(_,[],_).
safe_queen(Q,[Q2|R],D):-Q=\=Q2,abs(Q-Q2)=\=D,D1 is D+1,safe_queen(Q,R,D1).
Colourising buffer ... done, 0.00 seconds, 69 fragments
Line: 1


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24 ?- % c:/users/harshita singh/onedrive/documents/practical-1 (knowledge base) compiled 0
.00 sec, -1 clauses
24 ?- enrolled(rahul, python).
true.
25 ?- enrolled(amit, dbms).
false.
26 ?-
Practical-1 (Knowledge Base).pl
File Edit Browse Compile Prolog Help
Practical-1 (Knowledge Base).pl
enrolled(rahul, python).
enrolled(rahul, ai).
enrolled(priya, java).
enrolled(priya, python).
enrolled(amit, ai).
enrolled(amit, java).
enrolled(shubham,dbms).
enrolled(shubham,java).
c:/users/harshita singh/onedrive/documents/practical-1 (knowledge base).pl compiled
Line: 1


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For built-in help, use ?- help(Topic). or ?- apropos(Word).
26 ?- add(5,10,A).
A = 15.
27 ?- subtract(8,6,H).
H = 2.
28 ?- multiply(4,7,D).
D = 28.
29 ?- divide(8,2,K).
K = 4.
30 ?- factorial(6,Y).
Y = 720 .
31 ?- fibonacci(7,U).
U = 13
Practical-2 Arithmetic operations and Fibonacci.pl
File Edit Browse Compile Prolog Help
Practical-2 Arithmetic operations and Fibonacci.pl
add(A,B,R):-R is A+B.
subtract(A,B,R):-R is A-B.
multiply(A,B,R):-R is A*B.
divide(A,B,R):-B=\=0,R is A/B.
factorial(0,1).
factorial(N,F):-N>0,N1 is N-1,factorial(N1,F1),F is N*F1.
fibonacci(0,0). fibonacci(1,1).
fibonacci(N,F):-N>1,N1 is N-1,N2 is N-2,fibonacci(N1,F1),fibonacci(N2,F2),F is F1+F2.
c:/users/harshita singh/onedrive/documents/practical-2 arithmetic operations and fibonacci.pl compiled
Line: 1

SWI-Prolog console

File Settings Tools Debug Help

```
32 ?- append_list([1,2],[3,4],X).
X = [1, 2, 3, 4].

33 ?- member_element(3,[1,2,3,4,5]).
true .

34 ?- member_element(8,[1,2,3,4,5]).
false.

35 ?- list_length([10,20,30,40,50],N).
N = 5.

36 ?- reverse_list([1,2,3,4,5],R).
R = [5, 4, 3, 2, 1].

37 ?-
```

Practical-3 List Operations.pl

File Edit Browse Compile Prolog Help

Practical-2 Arithmetic operations and Fibonacci.pl | Practical-3 List Operations.pl

```
append_list([],L,L).
append_list([H|T],L,[H|R]):-append_list(T,L,R).
member_element(X,[X|_]).
member_element(X,[_|T]):-member_element(X,T).
list_length([],0).
list_length([_|T],N):-list_length(T,N1),N is N1+1.
reverse_list([],[]).
reverse_list([H|T],R):-reverse_list(T,RT),append_list(RT,[H],R).
```

Colourising buffer ... done, 0.01 seconds, 52 fragments Line: 1

SWI-Prolog console

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```
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37 ?- find_class(monday,9,S,F,R).
S = java,
F = sourav_sir,
R = room101.

38 ?- find_class(monday,11,S,F,R).
S = ai,
F = rahul_sir,
R = room102.

39 ?- faculty_subject(F,dsa).
F = neha_maam.

40 ?- faculty_subject(sourav_sir,S).
S = java.

41 ?-
```

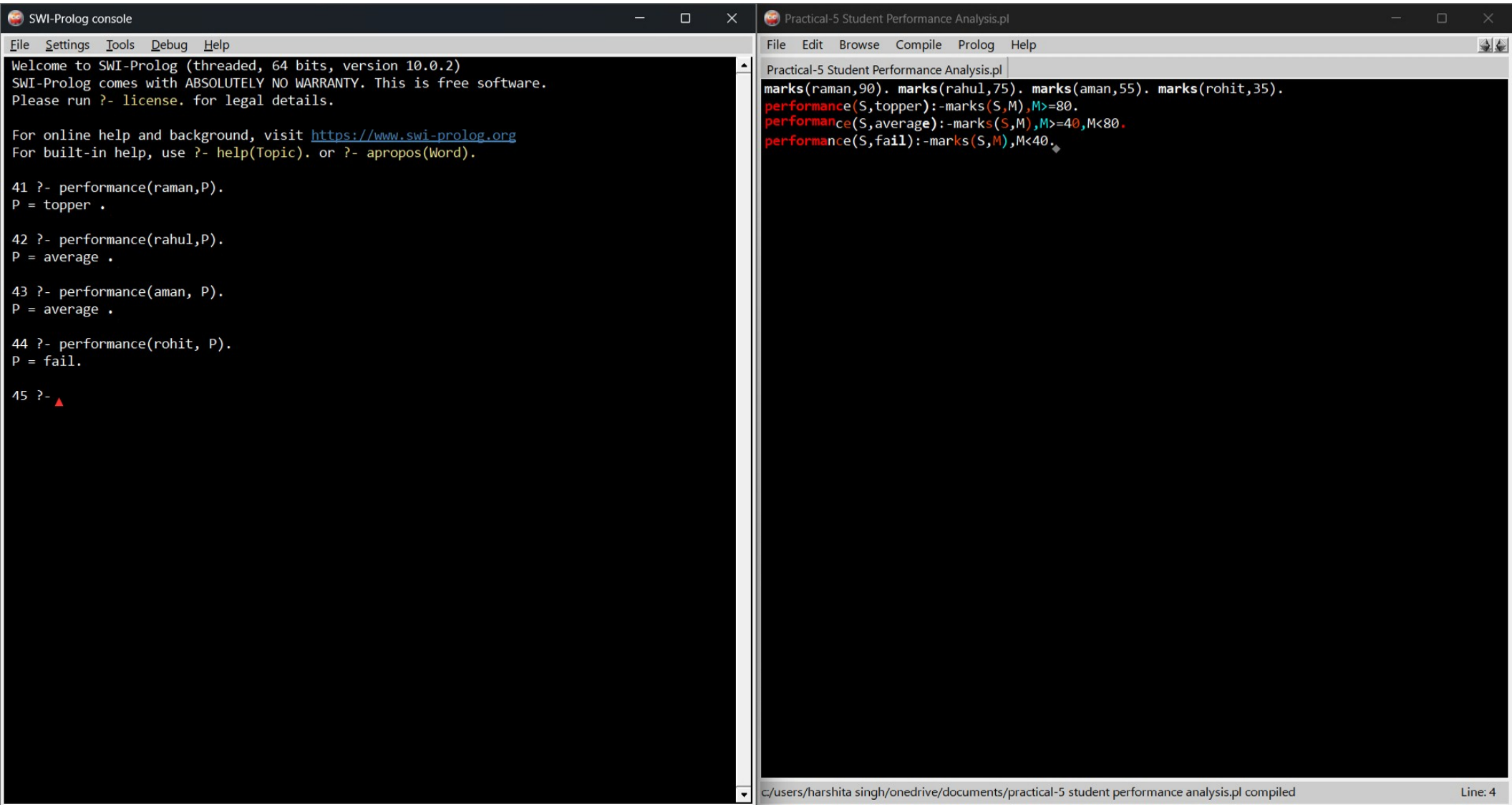
Practical-4 College Timetable.pl

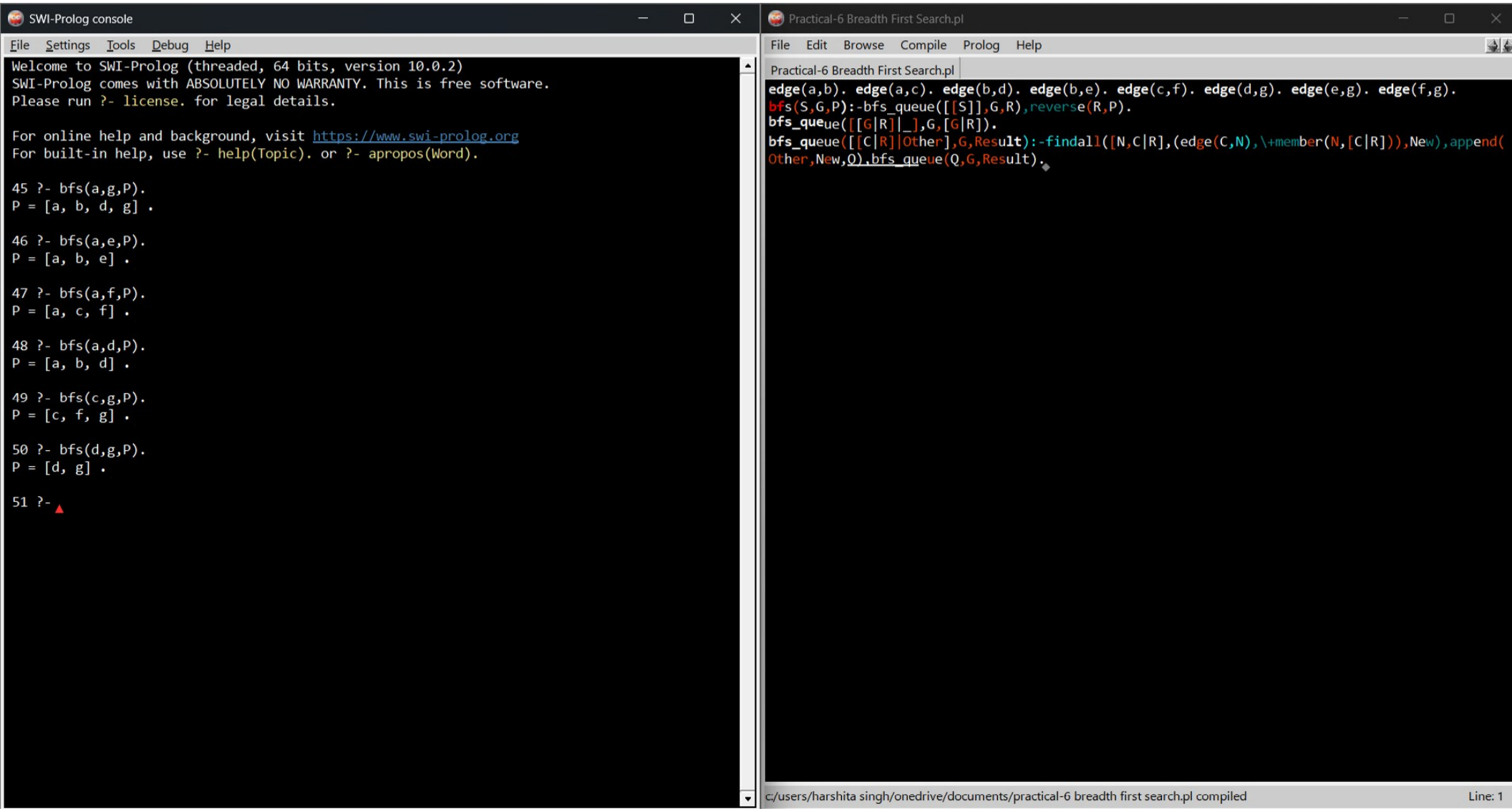
File Edit Browse Compile Prolog Help

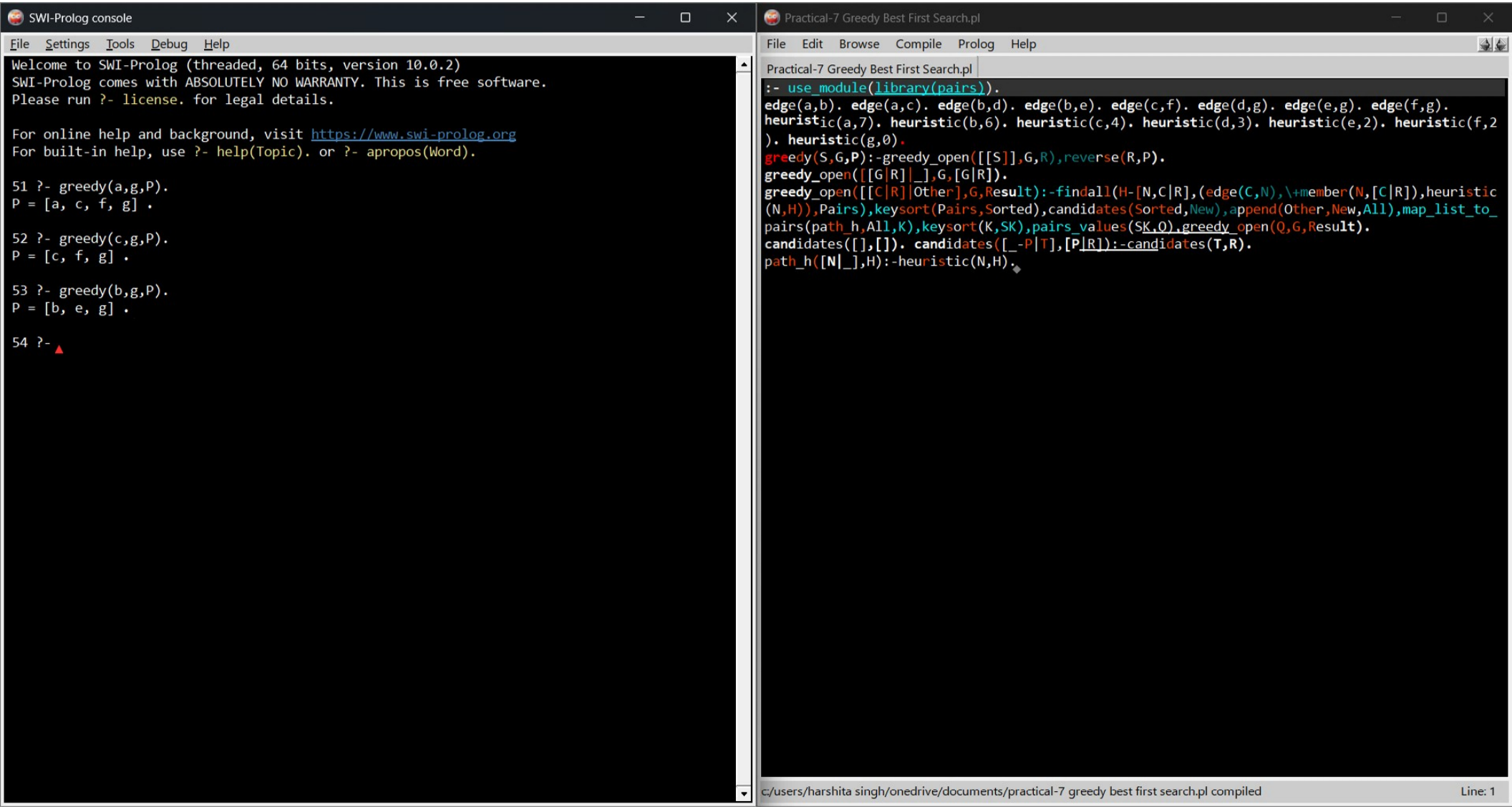
Practical-4 College Timetable.pl

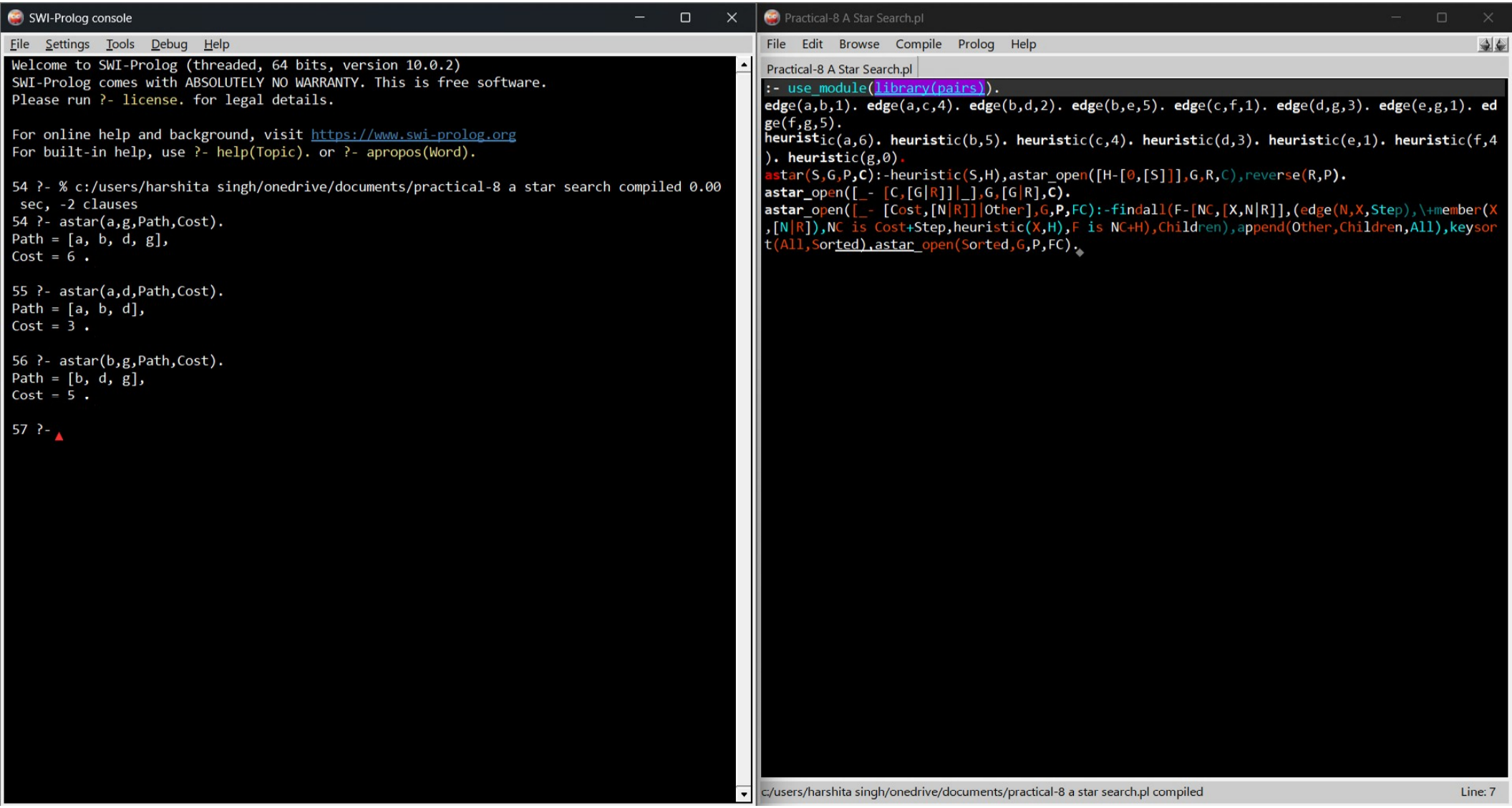
```
class(monday,9,java,sourav_sir,room101).
class(monday,11,ai,rahul_sir,room102).
class(tuesday,9,dbms,amit_sir,room103).
class(wednesday,10,dsa,neha_maam,room104).
find_class(D,T,S,F,R):-class(D,T,S,F,R).
faculty_subject(F,S):-class(_,_,S,F,_).
```

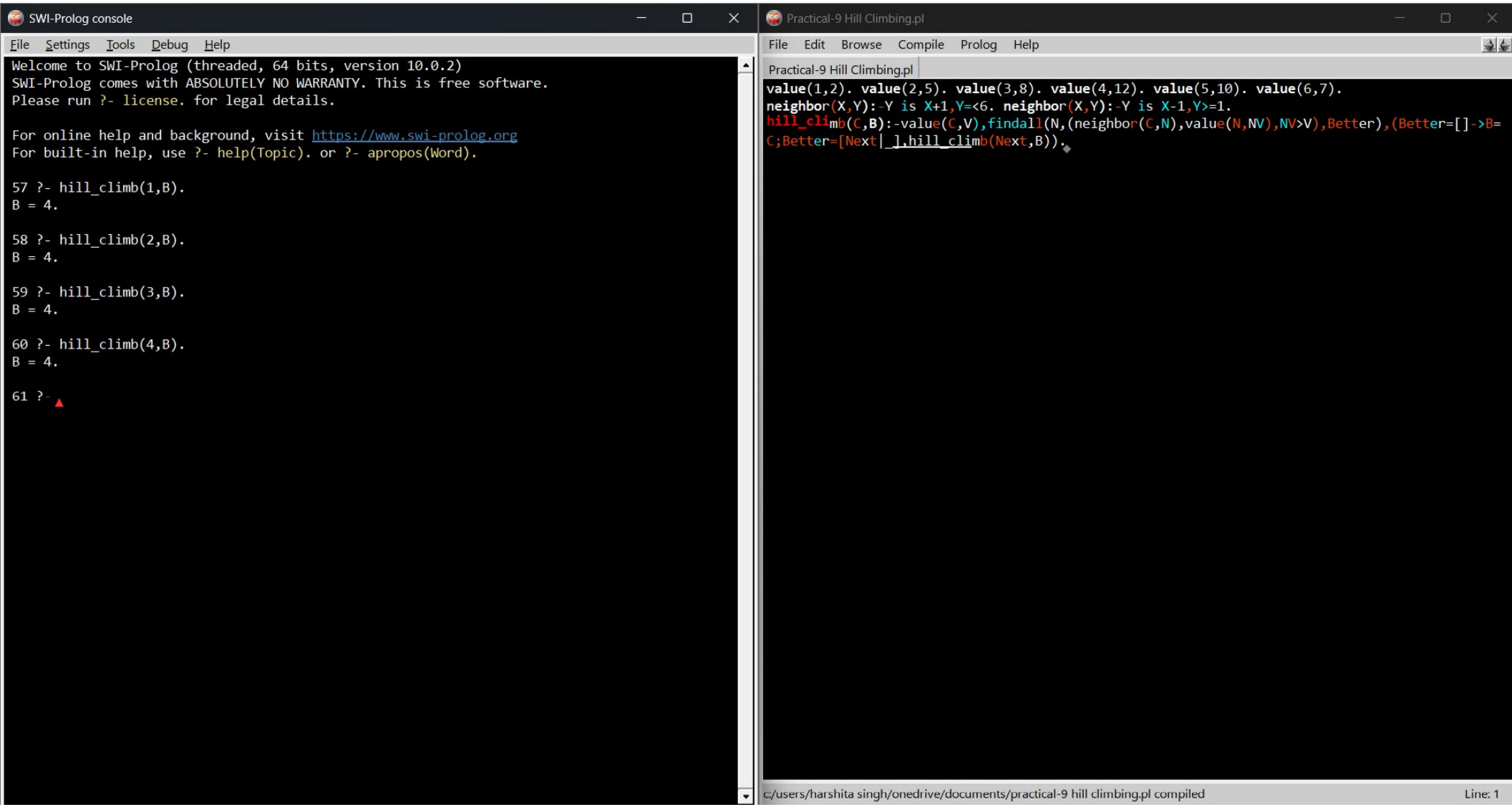
c:/users/harshita singh/onedrive/documents/practical-4 college timetable.pl compiled Line: 1

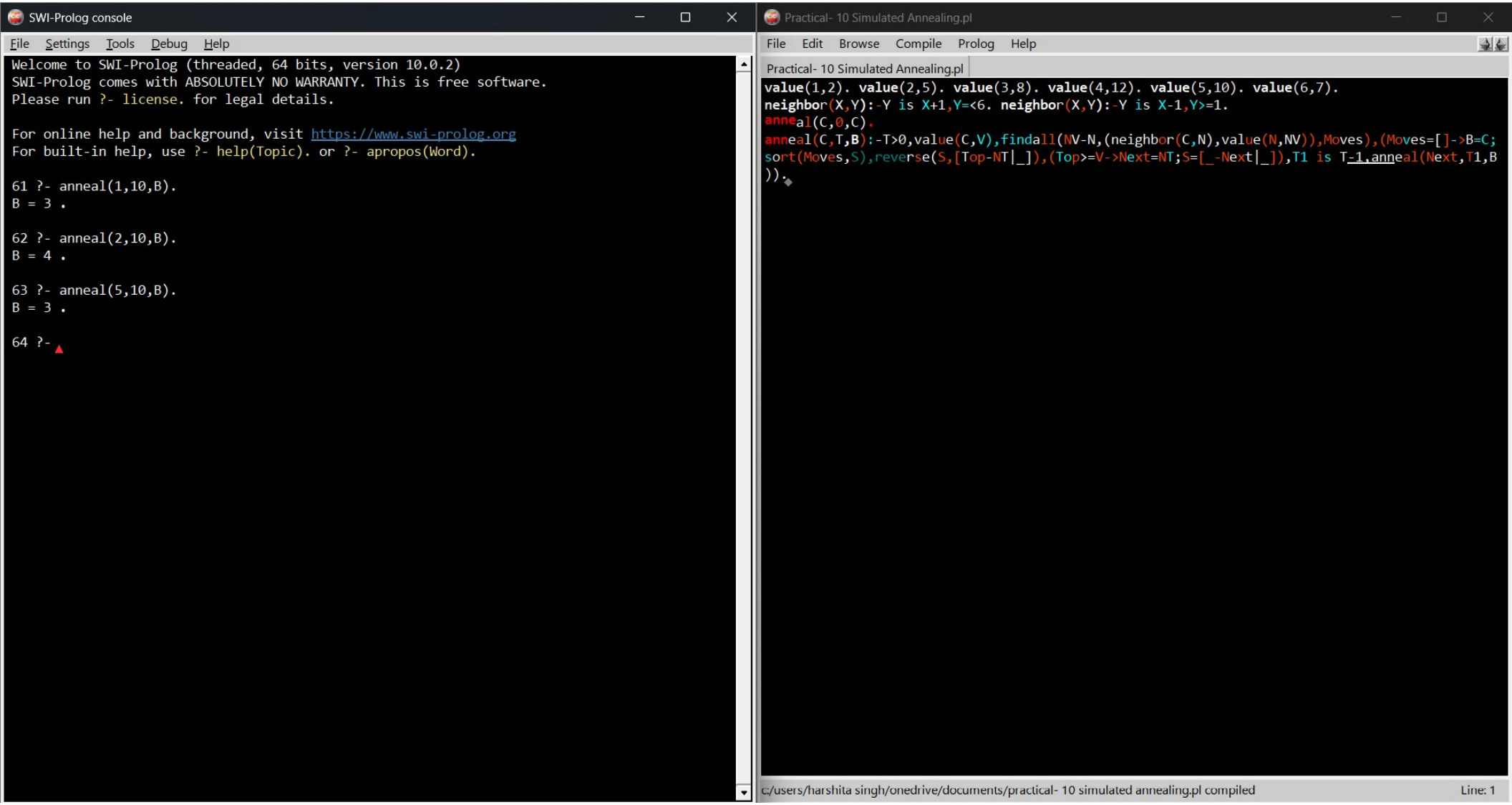

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For built-in help, use ?- help(Topic). or ?- apropos(Word).
41 ?- performance(raman,P).
P = topper .
42 ?- performance(rahul,P).
P = average .
43 ?- performance(aman, P).
P = average .
44 ?- performance(rohit, P).
P = fail.
45 ?-
Practical-5 Student Performance Analysis.pl
File Edit Browse Compile Prolog Help
Practical-5 Student Performance Analysis.pl
marks(raman,90). marks(rahul,75). marks(aman,55). marks(rohit,35).
performance(S,topper):-marks(S,M),M>=80.
performance(S,average):-marks(S,M),M>=40,M<80.
performance(S,fail):-marks(S,M),M<40.
c:/users/harshita singh/onedrive/documents/practical-5 student performance analysis.pl compiled
Line: 4

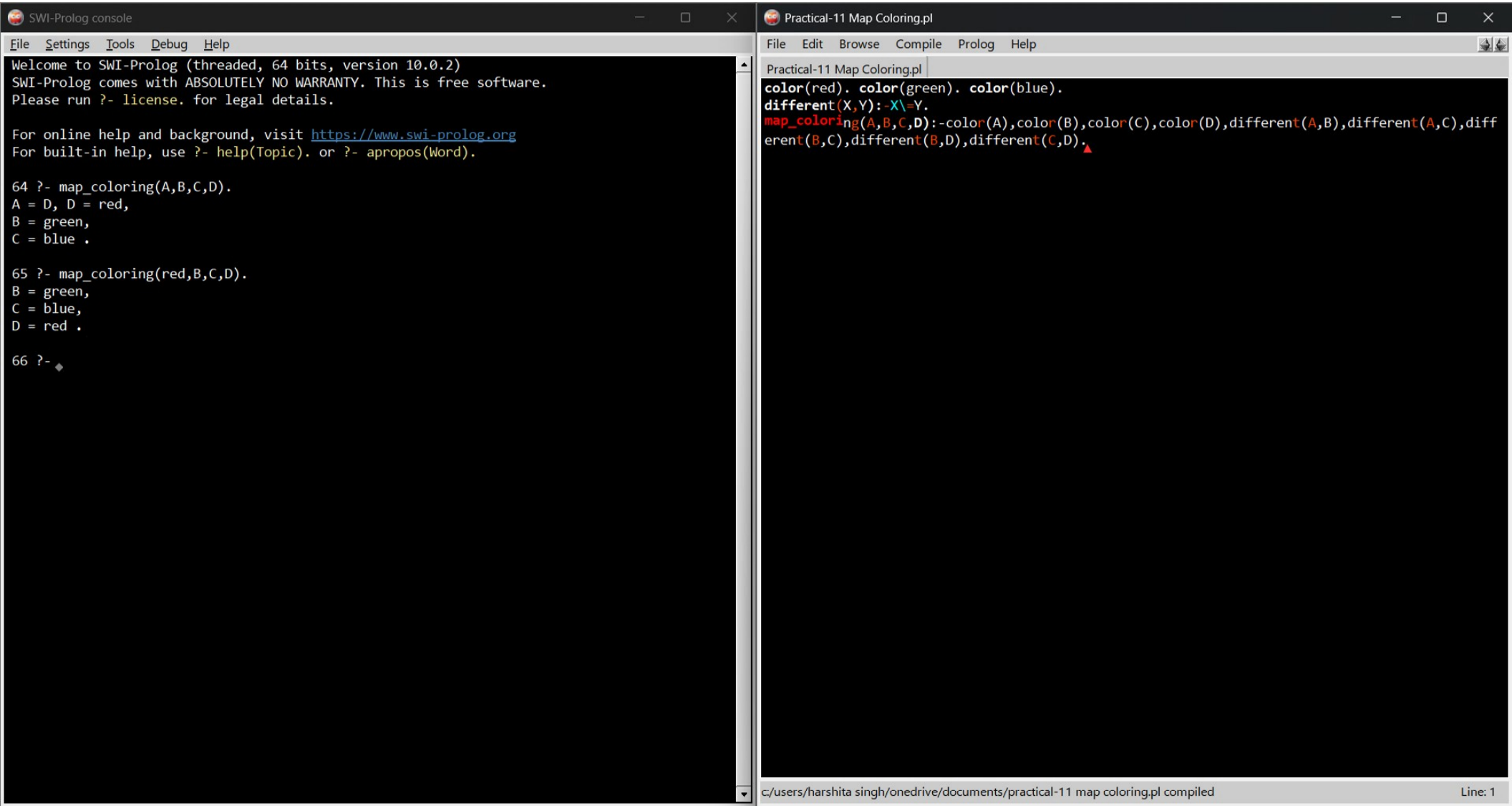

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For built-in help, use ?- help(Topic). or ?- apropos(Word).
45 ?- bfs(a,g,P).
P = [a, b, d, g] .
46 ?- bfs(a,e,P).
P = [a, b, e] .
47 ?- bfs(a,f,P).
P = [a, c, f] .
48 ?- bfs(a,d,P).
P = [a, b, d] .
49 ?- bfs(c,g,P).
P = [c, f, g] .
50 ?- bfs(d,g,P).
P = [d, g] .
51 ?-
Practical-6 Breadth First Search.pl
File Edit Browse Compile Prolog Help
Practical-6 Breadth First Search.pl
edge(a,b). edge(a,c). edge(b,d). edge(b,e). edge(c,f). edge(d,g). edge(e,g). edge(f,g).
bfs(S,G,P):-bfs_queue([[S]],G,R),reverse(R,P).
bfs_queue([[G|R]|_],G,[G|R]).
bfs_queue([[C|R]|Other],G,Result):-findall([N,C|R],(edge(C,N),\+member(N,[C|R])),New),append(
Other,New,Q),bfs_queue(Q,G,Result).
c:/users/harshita singh/onedrive/documents/practical-6 breadth first search.pl compiled
Line: 1


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For built-in help, use ?- help(Topic). or ?- apropos(Word).
51 ?- greedy(a,g,P).
P = [a, c, f, g] .
52 ?- greedy(c,g,P).
P = [c, f, g] .
53 ?- greedy(b,g,P).
P = [b, e, g] .
54 ?-
Practical-7 Greedy Best First Search.pl
File Edit Browse Compile Prolog Help
Practical-7 Greedy Best First Search.pl
:- use_module(library(pairs)).
edge(a,b). edge(a,c). edge(b,d). edge(b,e). edge(c,f). edge(d,g). edge(e,g). edge(f,g).
heuristic(a,7). heuristic(b,6). heuristic(c,4). heuristic(d,3). heuristic(e,2). heuristic(f,2
). heuristic(g,0).
greedy(S,G,P):-greedy_open([[S]],G,R),reverse(R,P).
greedy_open([[G|R]|_],G,[G|R]).
greedy_open([[C|R]|Other],G,Result):-findall(H-[N,C|R],(edge(C,N),\+member(N,[C|R]),heuristic
(N,H)),Pairs),keysort(Pairs,Sorted),candidates(Sorted,New),append(Other,New,All),map_list_to_
pairs(path_h,All,K),keysort(K,SK),pairs_values(SK,O),greedy_open(Q,G,Result).
candidates([],[]). candidates([_-P|T],[P|R]):-candidates(T,R).
path_h([N|_],H):-heuristic(N,H).
c:/users/harshita singh/onedrive/documents/practical-7 greedy best first search.pl compiled
Line: 1


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For built-in help, use ?- help(Topic). or ?- apropos(Word).
54 ?- % c:/users/harshita singh/onedrive/documents/practical-8 a star search compiled 0.00 sec, -2 clauses
54 ?- astar(a,g,Path,Cost).
Path = [a, b, d, g],
Cost = 6 .
55 ?- astar(a,d,Path,Cost).
Path = [a, b, d],
Cost = 3 .
56 ?- astar(b,g,Path,Cost).
Path = [b, d, g],
Cost = 5 .
57 ?-
Practical-8 A Star Search.pl
File Edit Browse Compile Prolog Help
Practical-8 A Star Search.pl
:- use_module(library(pairs)).
edge(a,b,1). edge(a,c,4). edge(b,d,2). edge(b,e,5). edge(c,f,1). edge(d,g,3). edge(e,g,1). edge(f,g,5).
heuristic(a,6). heuristic(b,5). heuristic(c,4). heuristic(d,3). heuristic(e,1). heuristic(f,4). heuristic(g,0).
astar(S,G,P,C):-heuristic(S,H),astar_open([H-[0,[S]]],G,R,C),reverse(R,P).
astar_open([_- [C,[G|R]]|_],G,[G|R],C).
astar_open([_- [Cost,[N|R]]|Other],G,P,FC):-findall(F-[NC,[X,N|R]],(edge(N,X,Step),\+member(X,[N|R]),NC is Cost+Step,heuristic(X,H),F is NC+H),Children),append(Other,Children,All),keysort(All,Sorted),astar_open(Sorted,G,P,FC).
c:/users/harshita singh/onedrive/documents/practical-8 a star search.pl compiled
Line: 7


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57 ?- hill_climb(1,B).
B = 4.
58 ?- hill_climb(2,B).
B = 4.
59 ?- hill_climb(3,B).
B = 4.
60 ?- hill_climb(4,B).
B = 4.
61 ?-
Practical-9 Hill Climbing.pl
File Edit Browse Compile Prolog Help
Practical-9 Hill Climbing.pl
value(1,2). value(2,5). value(3,8). value(4,12). value(5,10). value(6,7).
neighbor(X,Y):-Y is X+1,Y=<6. neighbor(X,Y):-Y is X-1,Y>=1.
hill_climb(C,B):-value(C,V),findall(N,(neighbor(C,N),value(N,NV),NV>V),Better),(Better=[]->B=C;Better=[Next|_],hill_climb(Next,B)).
c:/users/harshita singh/onedrive/documents/practical-9 hill climbing.pl compiled
Line: 1


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For built-in help, use ?- help(Topic). or ?- apropos(Word).
61 ?- anneal(1,10,B).
B = 3 .
62 ?- anneal(2,10,B).
B = 4 .
63 ?- anneal(5,10,B).
B = 3 .
64 ?-
Practical- 10 Simulated Annealing.pl
File Edit Browse Compile Prolog Help
Practical- 10 Simulated Annealing.pl
value(1,2). value(2,5). value(3,8). value(4,12). value(5,10). value(6,7).
neighbor(X,Y):-Y is X+1,Y=<6. neighbor(X,Y):-Y is X-1,Y>=1.
anneal(C,0,C).
anneal(C,T,B):-T>0,value(C,V),findall(NV-N,(neighbor(C,N),value(N,NV)),Moves),(Moves=[]->B=C;
sort(Moves,S),reverse(S,[Top-NT|_]),(Top>=V->Next=NT;S=[_-Next|_]),T1 is T-1,anneal(Next,T1,B
)).
c:/users/harshita singh/onedrive/documents/practical- 10 simulated annealing.pl compiled
Line: 1


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For built-in help, use ?- help(Topic). or ?- apropos(Word).
64 ?- map_coloring(A,B,C,D).
A = D, D = red,
B = green,
C = blue .
65 ?- map_coloring(red,B,C,D).
B = green,
C = blue,
D = red .
66 ?-
Practical-11 Map Coloring.pl
File Edit Browse Compile Prolog Help
Practical-11 Map Coloring.pl
color(red). color(green). color(blue).
different(X,Y):-X\=Y.
map_coloring(A,B,C,D):-color(A),color(B),color(C),color(D),different(A,B),different(A,C),different(B,C),different(B,D),different(C,D).
c:/users/harshita singh/onedrive/documents/practical-11 map coloring.pl compiled
Line: 1